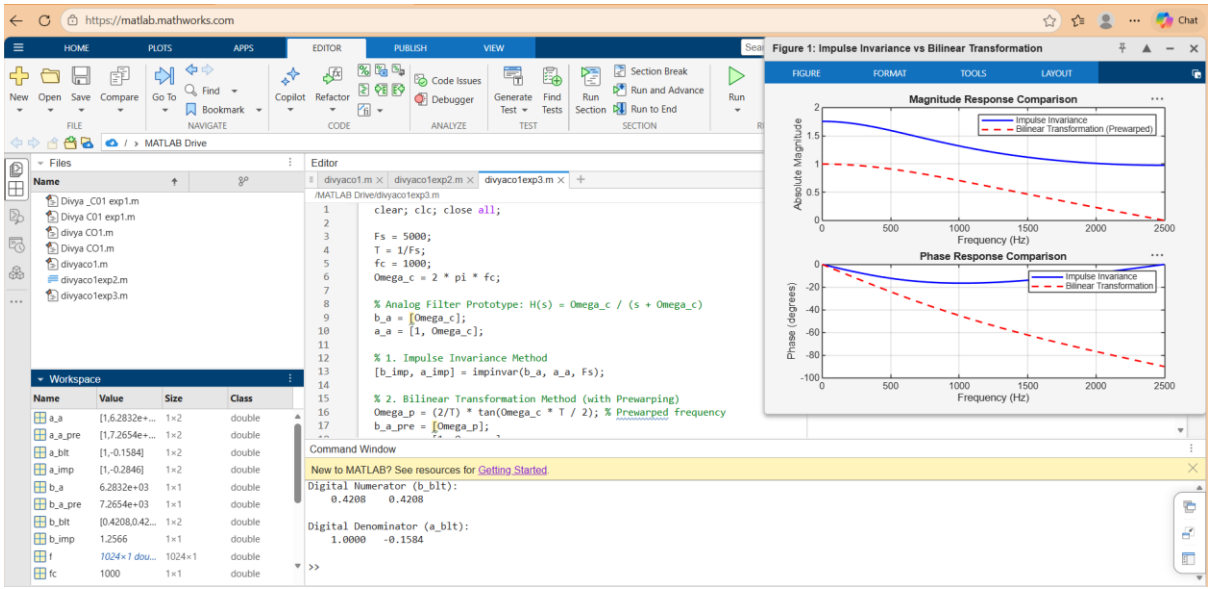

https://matlab.mathworks.com
Chat
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
FILE
NAVIGATE
CODE
ANALYZE
TEST
SECTION
MATLAB Drive
Files
Name
Divya _C01 exp1.m
Divya C01 exp1.m
divya CO1.m
Divya CO1.m
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
Workspace
Name
Value
Size
Class
a_a
[1,6.2832e+...
1×2
double
a_a_pre
[1,7.2654e+...
1×2
double
a_blt
[1,-0.1584]
1×2
double
a_imp
[1,-0.2846]
1×2
double
b_a
6.2832e+03
1×1
double
b_a_pre
7.2654e+03
1×1
double
b_blt
[0.4208,0.42...
1×2
double
b_imp
1.2566
1×1
double
f
1024×1 dou...
1024×1
double
fc
1000
1×1
double
Editor
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
/MATLAB Drive/divyaco1exp3.m
clear; clc; close all;
Fs = 5000;
T = 1/Fs;
fc = 1000;
Omega_c = 2 * pi * fc;
% Analog Filter Prototype: H(s) = Omega_c / (s + Omega_c)
b_a = [Omega_c];
a_a = [1, Omega_c];
% 1. Impulse Invariance Method
[b_imp, a_imp] = impinvar(b_a, a_a, Fs);
% 2. Bilinear Transformation Method (with Prewarping)
Omega_p = (2/T) * tan(Omega_c * T / 2); % Prewarped frequency
b_a_pre = [Omega_p];
Command Window
New to MATLAB? See resources for Getting Started.
Digital Numerator (b_blt):
0.4208 0.4208
Digital Denominator (a_blt):
1.0000 -0.1584
>>
Figure 1: Impulse Invariance vs Bilinear Transformation
FIGURE
FORMAT
TOOLS
LAYOUT
Magnitude Response Comparison
Impulse Invariance
Bilinear Transformation (Prewarped)
Absolute Magnitude
Frequency (Hz)
Phase Response Comparison
Impulse Invariance
Bilinear Transformation
Phase (degrees)
Frequency (Hz)